

Problem 18.19

What is the present value of receiving \$30 at the end of each year forever, starting 9 years from now? Assume an annual rate of interest of 7%.

Solution

The payments form a deferred perpetuity-immediate. Since the first payment is received at the end of year 9, the value one year before the first payment, which is at time 8, is

$$30a_{\infty|} = 30 \left(\frac{1}{i} \right).$$

With $i = 0.07$,

$$30a_{\infty|} = 30 \left(\frac{1}{0.07} \right).$$

To find the present value at time 0, discount this value for 8 years:

$$PV = 30 \left(\frac{1}{0.07} \right) (1.07)^{-8}.$$

Thus,

$$PV = 249.4324734.$$

Therefore, the present value is

$$\boxed{249.43}.$$